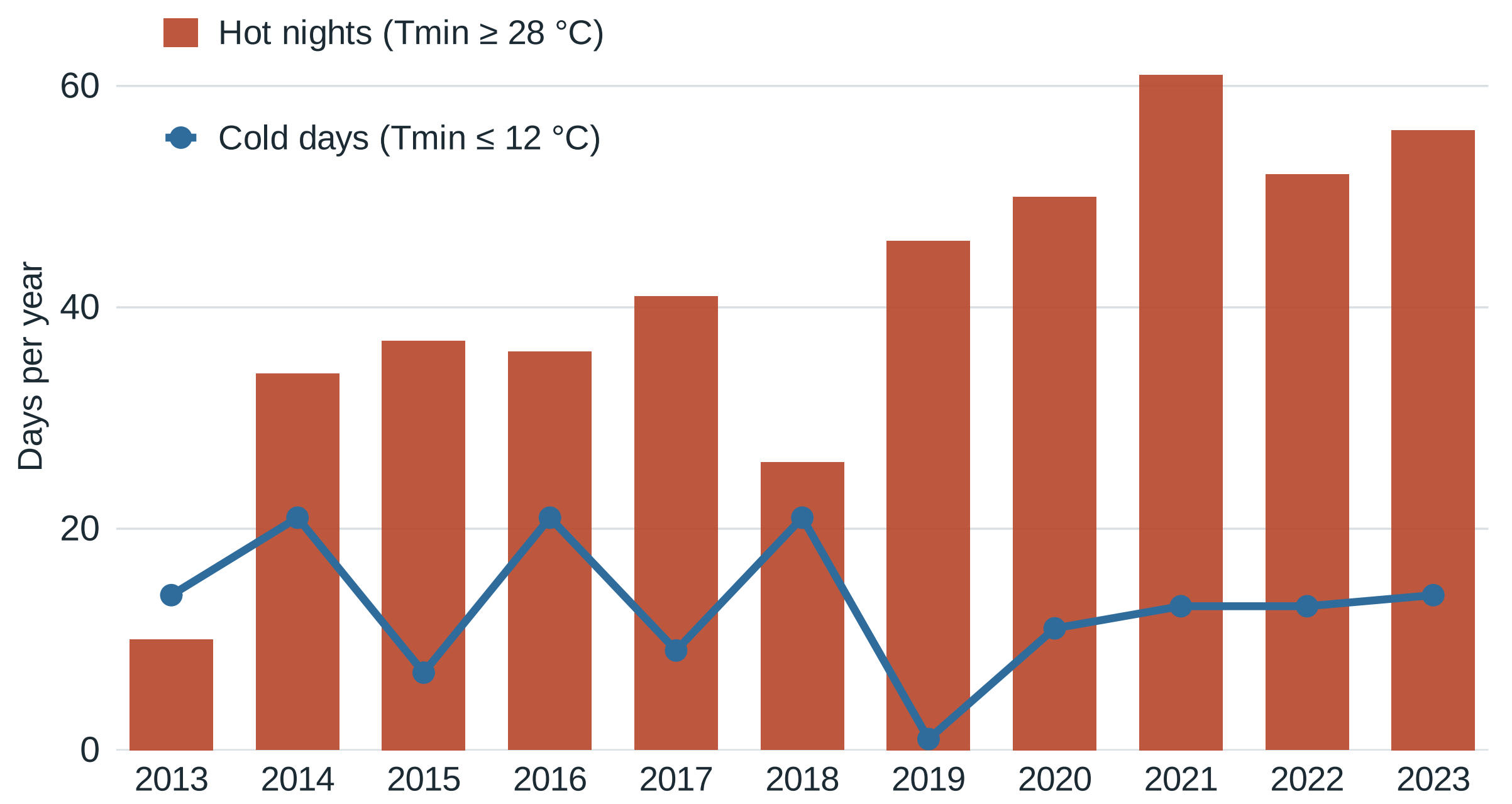


INTRODUCTION

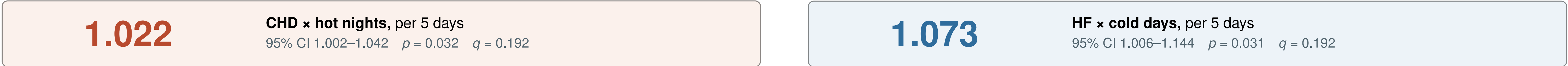
Hong Kong is warming fastest at night: hot nights at the Hong Kong Observatory rose from 10 days in 2013 to a peak of 61 in 2021, while winter cold days have not disappeared. For the growing number of people living with diabetes or hypertension — a group at high cardiovascular risk — adaptation planning must budget for the hospital burden, not only mortality. Whether monthly heat and cold burdens track first cardiovascular hospitalisations in this group has not been quantified.



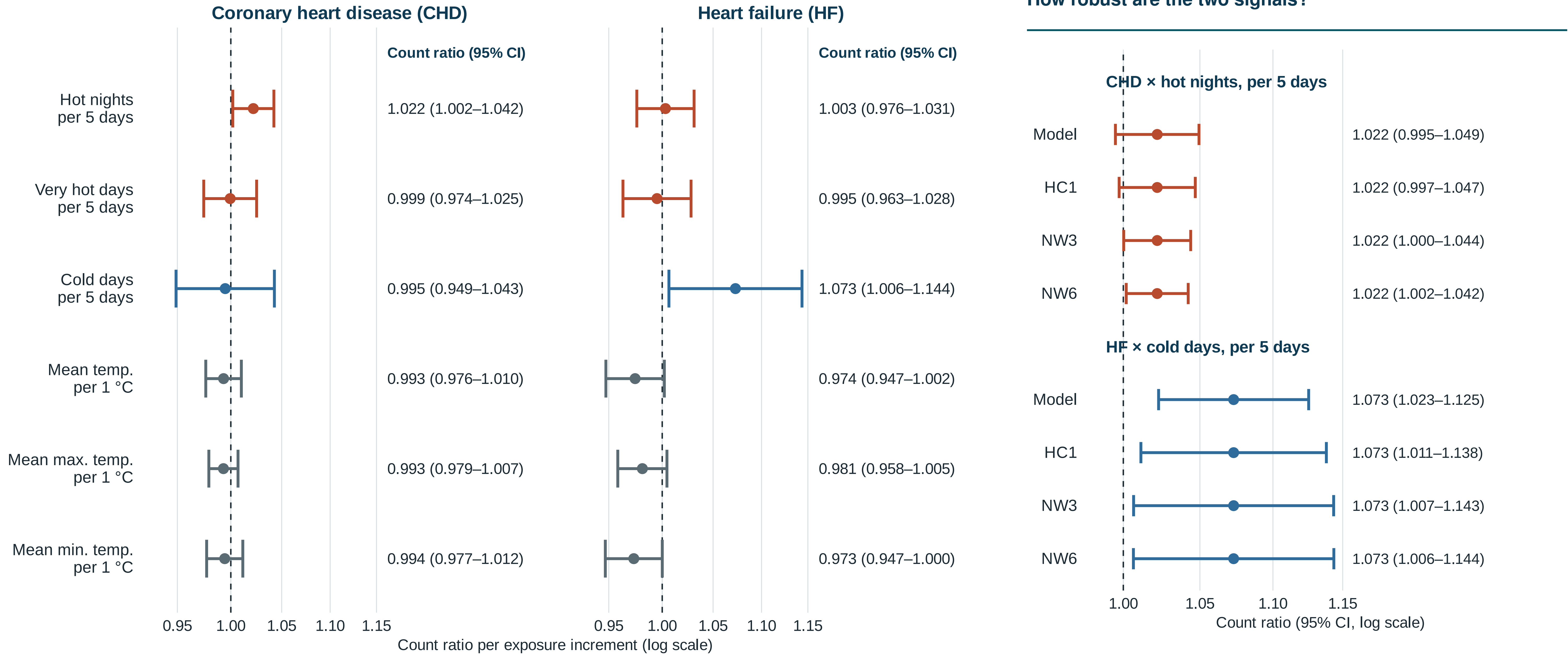
Official Hong Kong Observatory extremes at Headquarters. Hot nights rose about fivefold over the decade; cold days recurred every winter — the exposure setting for this study.

RESULTS

Twelve pre-specified models, read as one panel — two contrasts stand out.



- **CHD x hot nights:** about **2.2% higher** monthly count per five additional hot nights — but the interval *includes 1* under two of four standard-error methods, so the signal is uncertainty-sensitive.
- **HF x cold days:** about **7.3% higher** monthly count per five additional cold days — the interval excludes 1 under all four methods.
- The other ten contrasts sit near 1. **All 12 Benjamini–Hochberg q -values exceed 0.19:** after correction for 12 comparisons, both signals remain exploratory.



The 12-model core panel: count ratios from single-exposure negative-binomial models (days-in-month offset); 95% intervals with Newey–West lag-6 standard errors. Values right of 1 mean more first hospitalisations in months with higher exposure. All 12 Benjamini–Hochberg $q > 0.19$.

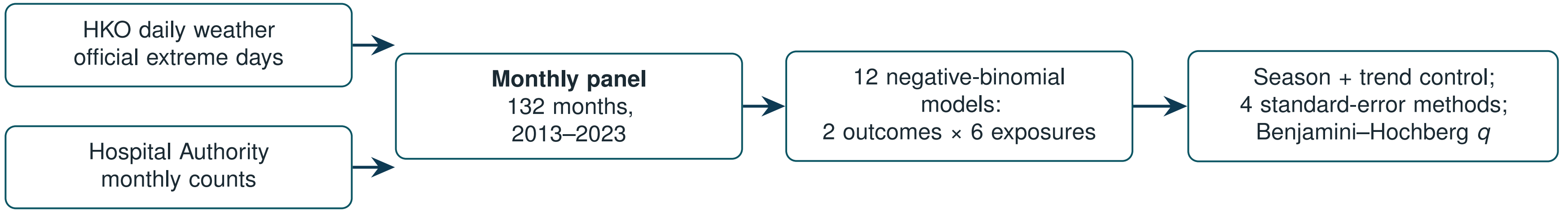
INTERPRETATION / LIMITATIONS

- All 12 q -values exceed 0.19 — these are **exploratory signals**, not confirmed effects.
- The CHD hot-night estimate is **sensitive to how uncertainty is computed**; the HF cold-day estimate is stable across methods.
- Monthly, territory-level data with no recorded admission cause support **no causal or principal-diagnosis claim**; a stroke extract was not delivered, so stroke is not reported.

OBJECTIVE

Are monthly temperature and extreme-day burdens associated with first **coronary heart disease (CHD)** or **heart failure (HF)** hospitalisations in a high-risk Hong Kong cohort?

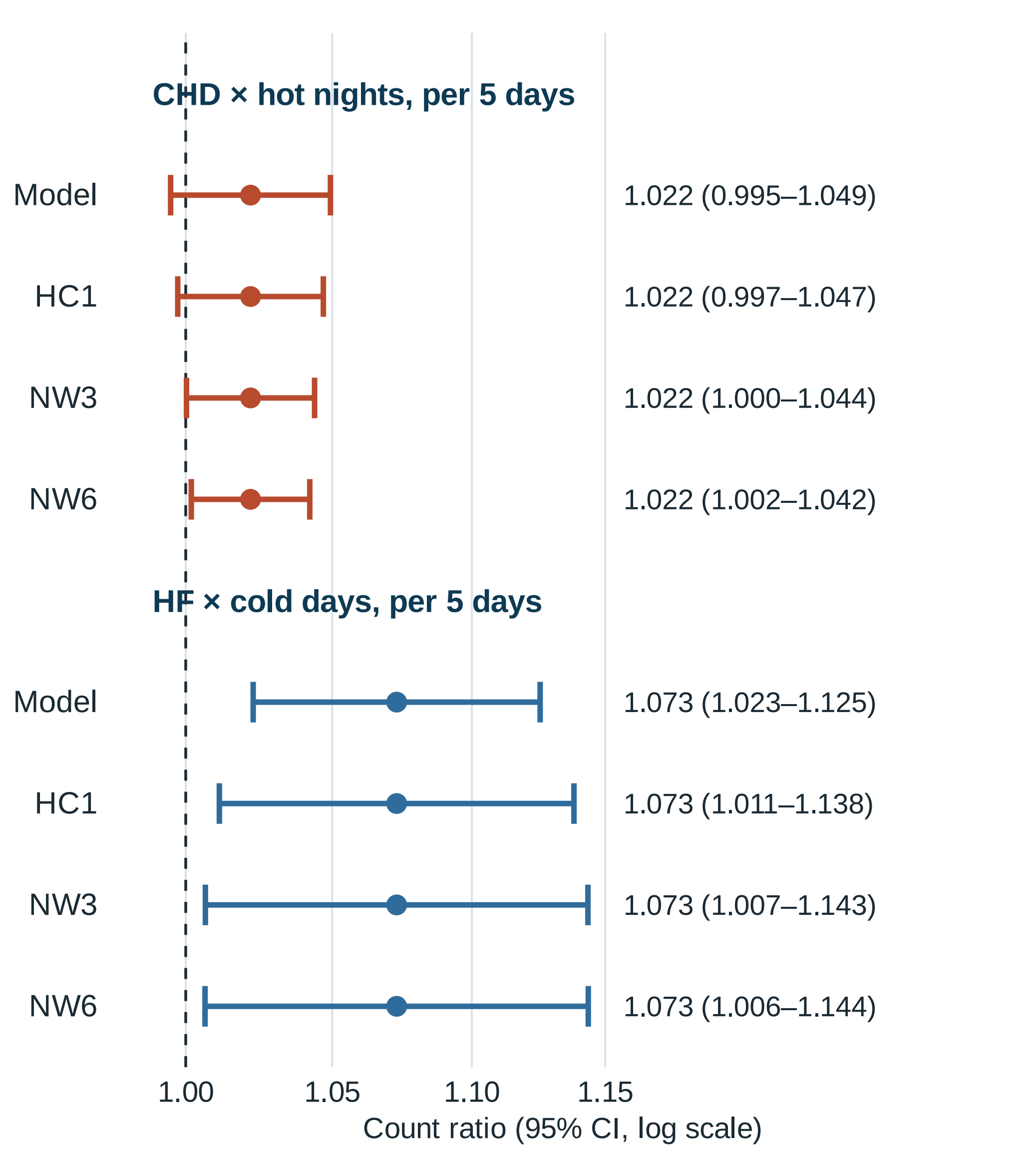
METHODS



- **Window and weather:** 132 territory-months, Jan 2013 – Dec 2023; daily records and official extreme-day definitions from the Hong Kong Observatory.
- **Outcomes:** governed Hospital Authority aggregate counts of *first hospitalisation after a first recorded CHD or HF diagnosis* in people with type 2 diabetes and/or hypertension — 156,156 CHD and 29,681 HF events. Admission cause is not recorded.
- **Models:** 2 outcomes x 6 exposures = 12 pre-specified negative-binomial regressions with calendar-month indicators, a trend spline, and a days-in-month offset.
- **Uncertainty:** four standard-error methods per contrast; Benjamini–Hochberg q across all 12 comparisons.

Definitions. Hot night: minimum 28 °C or higher; very hot day: maximum 33 °C or higher; cold day: minimum 12 °C or lower (HKO definitions). Estimates are count ratios per 5 extreme days or per 1 °C.

How robust are the two signals?



The two leading contrasts under four standard-error methods. The CHD hot-night interval includes 1 under Model and HC1; the HF cold-day interval excludes 1 under all four. The pre-specified analysis of record uses NW6; methods were not selected by interval position.

CONCLUSION

Monthly data suggest **different thermal patterns** for CHD and HF first hospitalisations — hot nights for CHD, cold days for HF — but the evidence is **exploratory, not confirmatory**: no contrast survived correction for twelve comparisons. The project leaves a **reproducible monthly framework** and a clear next requirement: governed **daily or cohort data with recorded admission cause**.

What would strengthen this: linked daily weather and admission records with cause of admission, so the monthly burden patterns seen here can be tested at the resolution at which heat and cold act.